Robust Literature Discovery from Minimal Seeds: Validating LitDiscover on APS Citation Benchmarks and Live Surveys

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Abstract

Automated literature discovery must do something retrieval cannot: recover a topic's literature from almost nothing. We introduce LitDiscover, a queue-driven engine that navigates citation graphs through three coordinated mechanisms — bidirectional traversal, Pareto-filtered hub suppression, and yield-gated continuation. Using the American Physical Society (APS) 2022 dataset (709,803 papers, 9,833,191 edges) as a closed benchmark, we evaluate recovery from minimal seed sets ($k \in \{1, 2, 3, 4, 5, 10\}$) against three landmark survey papers. LitDiscover achieves near-complete bibliography recovery from as few as one to five seeds, demonstrating that high-recall discovery is achievable under realistic minimal-seeding conditions. Residual misses are structurally peripheral — low in-degree papers that are adjacent to but below the traversal's productive horizon — not randomly distributed across the domain. Live validation on three open-domain surveys confirms that the same yield-governed control logic transfers outside the benchmark: yield collapse is a reliable operational signal that engages at a principled structural boundary, with domain density governing how far the coverage frontier extends.

Keywords: literature discovery, citation graph traversal, systematic review, snowballing, bibliographic retrieval

1 Introduction

The central practical problem in automated literature review is not merely how to summarize papers once they have been found, but how to **discover** a topic comprehensively from incomplete starting information. Real users rarely begin with a complete domain map or even a strong initial bibliography. More commonly, they begin with a query, a few candidate papers, and an uncertain boundary between relevant and irrelevant work. A useful review system must therefore solve a discovery problem under sparse initial supervision.

Citation graphs provide a natural substrate for this task because they expose relational structure that keyword search alone cannot recover (Redner, 2005). A paper's references reveal backward intellectual lineage, while incoming citations reveal later work that builds on or reacts to it. In principle, repeated traversal over this graph should allow a system to expand from a small local starting point into a much larger topical neighbourhood. In practice, however, naive traversal fails because citation networks are highly asymmetric (Barabási, 2016; Barabási and Albert, 1999). Backward traversal is naturally bounded by bibliography length, whereas forward traversal rapidly encounters generic high-degree hubs that connect many otherwise distant parts of the literature. Unconstrained exploration therefore turns a promising discovery process into an intractable screening problem.

LitDiscover addresses this problem as a bounded traversal architecture. Rather than treating literature discovery as open-ended graph expansion, it organises discovery around a queue-driven loop in which search, screening, traversal, and escape operations are all subordinated to a single yield-governed control policy. Three architectural ideas are central. First, the system uses **bidirectional traversal**, because neither backward nor forward expansion alone is sufficient to recover a topic. Second, it applies **Pareto-filtered forward expansion** to suppress high-degree hubs that disproportionately drive graph explosion and topic drift. Third, it uses a **yield-gated continuation rule** to determine when local expansion remains productive and when the system should instead jump to a fresh part of the graph.

This paper evaluates LitDiscover in the regime that matters most for deployment: **minimal seeding** ($k \in \{1, 2, 3, 4, 5, 10\}$). The evaluation uses APS as a closed citation universe and derives gold sets directly from published survey bibliographies, giving a precise and bias-free recall metric. The same configuration is then run against three live open-domain surveys via the Semantic Scholar API, testing whether the yield-governed control logic transfers outside the benchmark.

The paper makes four contributions. First, it gives a structural account of why bounded traversal is necessary: hub suppression and yield stopping are principled responses to power-law degree skew, not arbitrary heuristics. Second, it shows that LitDiscover achieves near-complete recall from as few as one to five seed papers — the regime real users actually occupy. Third, it characterises the residual gap through miss analysis, showing that what the system misses is structurally peripheral rather than randomly lost; this matters because it distinguishes a principled stopping boundary from architectural failure, and informs when relaxing the Pareto or yield parameters would actually help. Fourth, it confirms via live validation that yield collapse is an operational signal in deployment, not just an offline metric; this is non-trivial because the closed-corpus simulation cannot demonstrate that the stopping rule generalises to open citation graphs with incomplete indexing and temporal gaps.

2 Related Work

This paper sits at the intersection of three areas.

Citation-graph analysis has characterised degree distributions, clustering, and community structure in academic networks (Barabási and Albert, 1999; Price, 1965). Heavy-tailed in-degree is well documented (Redner, 2005; Barabási, 2016): a small fraction of papers mediate a disproportionate share of cross-domain connectivity, motivating the Pareto filter design. Recent LLM-based simulation work corroborates this mechanistically — Ji et al. (2025) find that the recommendation algorithm's tendency to surface highly cited papers is a stronger driver of preferential attachment than author citation bias, explaining how hubs self-reinforce over time.

Automated systematic review and evidence synthesis spans a spectrum from manual-heavy to fully automated. Early tools such as SWIFT-Review (Howard et al., 2016) and RobotSearch (Marshall and Wallace, 2019) combine keyword or ML-based screening with manual corpus assembly. Active-learning frameworks such as ASReview (van de Schoot et al., 2021) go further by prioritising which pre-assembled candidates to screen first, substantially reducing manual effort — but all of these assume the candidate pool is already given. LLM-assisted exploration tools (Elicit (Elicit, 2024), ResearchRabbit (ResearchRabbit, 2025)) extend this with query-driven discovery, but without a formal stopping criterion. LitDiscover sits at the missing end of this spectrum: it *constructs* the candidate pool via graph traversal before screening begins, governed by an explicit yield-based termination rule.

Graph compression and hub-aware sampling provides the theoretical grounding for the Pareto filter. Work on k-core decomposition (Batagelj and Zaversnik, 2003) and degree-constrained sampling (Stumpf et al., 2005) shows that removing high-degree hubs substantially reduces edge volume while preserving reachability. The Pareto filter is a soft variant of this idea applied to the forward traversal frontier.

These three areas converge in a way that directly motivates LitDiscover. Citation-graph analysis supplies the observation that degree skew is structural and predictable —

it is not a nuisance to be filtered out after the fact, but a property that should govern traversal design from the outset. Systematic review methodology supplies the recall objective: what matters is not sampling the literature but recovering it completely enough to support synthesis. Hub-aware sampling supplies the filter mechanism: the Pareto threshold is a soft analogue of core decomposition, applied not to compress a static graph but to constrain a live traversal frontier. No prior tool connects all three: snowballing tools lack principled stopping; systematic review tools assume a pre-assembled corpus; graph compression work does not address the cold-start or recall-framing concerns of literature discovery.

LitDiscover formalises and automates the snowballing methodology (Wohlin, 2014) — systematic bidirectional citation chaining from seed papers — replacing manual curation with a yield-governed queue and LLM screening. The distinguishing feature relative to prior citation-expansion tools (CiteSpace (Chen, 2006), Connected Papers (Connected Papers, 2025)) is the minimal-seed framing: those systems assume a large anchor set; LitDiscover bootstraps a domain bibliography from two to five entry points.

3 Architecture and Control Logic

LitDiscover is best understood as a queue-driven discovery loop rather than a static retrieval pipeline. At a high level, the system alternates between **search**, **screening**, **traversal**, and **escape** operations until it reaches a stable state. The logic can be summarised as:

`SEED → SEARCH → SCREEN → TRAVERSE → ESCAPE → STABLE`

The system begins from an initial seed set, which may consist of search results, user-supplied papers, or a mixture of the two. These papers enter a unified screening queue. Screening decisions then identify which items should be retained as relevant and therefore eligible for expansion. Expansion occurs through citation traversal, but only under bounded conditions.

The first architectural principle is **bidirectional traversal**. Backward traversal follows references made by included papers and is essential because relevant older work is often directly cited by on-topic seeds. Forward traversal follows papers that cite included items and is equally important because it exposes later developments that no initial bibliography can contain. Yet forward traversal is also the primary source of explosion. Foundational papers, methods papers, and generic benchmark papers accumulate citations across many subfields, so blindly following their incoming edges pushes the system into large off-topic regions of the graph.

The second architectural principle is therefore **Pareto-filtered forward expansion**. Rather than expanding equally through all forward neighbours, LitDiscover suppresses a high-degree tail of forward candidates. The rationale is graph-theoretic: in a heavy-tailed network (Barabási, 2016), a relatively small fraction of nodes mediate a disproportionate share of edges. Those nodes are often precisely the hubs that maximise breadth while minimising topical specificity. The Pareto filter is thus not only an efficiency device but also a topicality control.

The third architectural principle is **yield-gated continuation**. Expansion is not judged solely by how many new nodes it reaches, but by how many useful papers it produces relative to the screening effort required. In the benchmark framing, this means that local expansion should continue only when recent discoveries remain sufficiently fruitful; once yield collapses, the system should either declare local exhaustion or invoke an **Escape** step. Escape selects a new topical anchor from papers already recovered but not yet used as traversal seeds — in deployment, this is realised by a fresh keyword query against an external index (Semantic Scholar); in the closed-corpus benchmark, it selects from the accumulated recovered set. The selected anchor re-enters the traversal loop as a new seed, bridging structurally disconnected or weakly connected clusters that local expansion cannot reach.

This architecture formalises literature discovery as an alternating process of **local exploitation** and **global re-entry**. Traversal explores dense neighbourhoods already connected to the current frontier. The Escape step compensates for structural gaps, indexing limitations, and disconnected subclusters by reintroducing a search-based jump. The system therefore does not assume that a topic is perfectly recoverable by graph traversal alone.

For the purposes of this paper, the key methodological claim is not only that final recall is high, but that the **control mechanisms themselves work correctly**: yield collapse reliably marks the boundary of productive local expansion, and the Escape step successfully bridges disconnected clusters. Validating these behaviours — measuring at what depth yield collapses, whether Escape finds genuinely new territory, and whether the system terminates without parameter sensitivity — is as important as the aggregate recall figure, because it establishes that the architecture is principled rather than tuned.

4 Closed-Corpus Benchmark and Experimental Design

To evaluate the architecture under controlled conditions, we use the APS 2022 citation dataset, a closed citation universe containing 709,803 papers and 9,833,191 citation edges across American Physical Society journals (American Physical Society, 2022). The closed-corpus property is essential because it ensures that every traversal step remains inside the benchmark, allowing structural reachability, recall, and miss analysis to be defined precisely.

The ground-truth topical targets are derived from three highly cited survey papers published in *Reviews of Modern Physics* (Imada et al., 1998; Bloch et al., 2008; Ozawa et al., 2019). Each survey contributes a bibliography that functions as a gold set for the associated topic.

Table 1: APS benchmark surveys and gold sets.

Survey ID	Topic	Year	Gold References
S1	Metal-insulator transitions	1998	582
S2	Ultracold gases	2008	432
S3	Topological photonics	2019	387

The benchmark has two conceptual layers. The first is **structural motivation**, where the survey papers are used to analyse citation-graph asymmetry and motivate bounded traversal. The second is the **cold-start validation**, which simulates realistic user conditions by replacing direct survey seeding with much smaller initial seed sets.

Table 2: Experimental design for the cold-start validation.

Experimental factor	Setting
Seed size	$k \in \{1, 2, 3, 4, 5, 10\}$
Seed quality	Top-k, random, contaminated
Traversal mode	Bidirectional with Pareto-80 forward filter
Continuation rule	Yield-gated ($\tau = 0.05$) with Escape re-entry
Main outcomes	Recall, screened set size, per-round gain, residual misses

The contaminated setting captures a plausible operational scenario in which a user's initial search returns a mixture of relevant and irrelevant papers. A robust discovery architecture should not require the seed set to be perfectly curated.

5 Structural Motivation for Bounded Traversal

The need for bounded traversal follows directly from the structure of the APS citation graph. The graph is heavy-tailed in its in-degree distribution (Barabási, 2016; Barabási and Albert, 1999) and much more constrained in its out-degree distribution. A small number of papers accumulate very large citation counts, while the number of references made by a typical paper remains much more limited. This asymmetry creates a directional imbalance in traversal cost.

Backward traversal is naturally disciplined by bibliography length. If a paper cites 20 to 50 relevant predecessors, then expanding backward from it remains relatively local. Forward traversal is qualitatively different. A single paper of broad methodological importance may

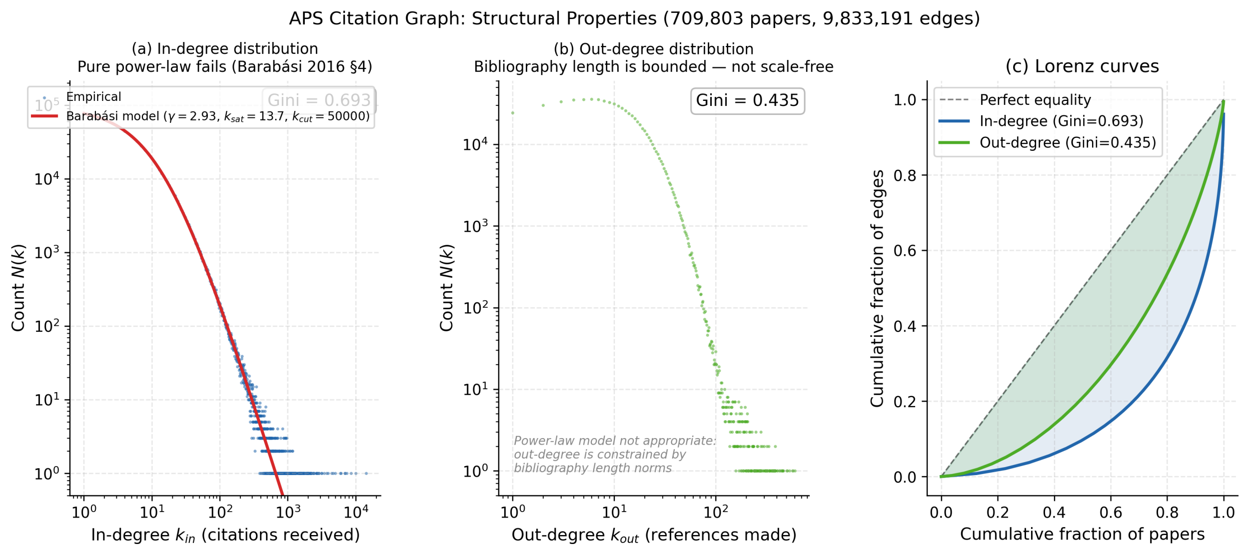

Figure 1: **Degree distributions of the APS citation graph.** In-degree (citations received) is heavy-tailed; out-degree (references made) is bounded. The asymmetry motivates why forward traversal explodes while backward traversal stays local.

be cited by hundreds or thousands of later papers that span multiple adjacent domains. Once traversal flows through such hubs, the search frontier ceases to represent a coherent topic.

This dynamic is visible in reachability experiments seeded from a small number of top-ranked gold-bibliography papers ($k = 5$, oracle seeding — upper bound on cold-start performance). Backward traversal reaches a substantially larger fraction of the gold set at each BFS depth than forward traversal, because the gold papers are densely interconnected through their shared references. Forward traversal recovers a complementary slice — particularly later work that cites the seeds — but also expands the reached set far faster, because high-degree citers act as bridges into adjacent domains. Bidirectional traversal achieves meaningfully higher gold-set coverage at every depth than either direction alone while growing the reached set more quickly than pure backward expansion.

This directional asymmetry has a mechanistic explanation. Empirical citation models find that approximately 80% of references arise from bibliographic copying — citing papers already cited by related works — rather than from independent global search (Goldberg et al., 2015). Papers within a coherent topical cluster therefore share a dense common reference layer, which backward traversal follows efficiently. Forward traversal enters that layer from the other side and immediately encounters the small fraction of highly cited works that bridge multiple subfields. These bridge nodes are precisely the hubs that forward traversal must cross to continue expanding, and crossing them pushes the frontier into off-topic regions. The Pareto filter is the operational response: suppressing the highest-degree forward candidates removes the inter-subfield bridges while preserving within-cluster reachability.

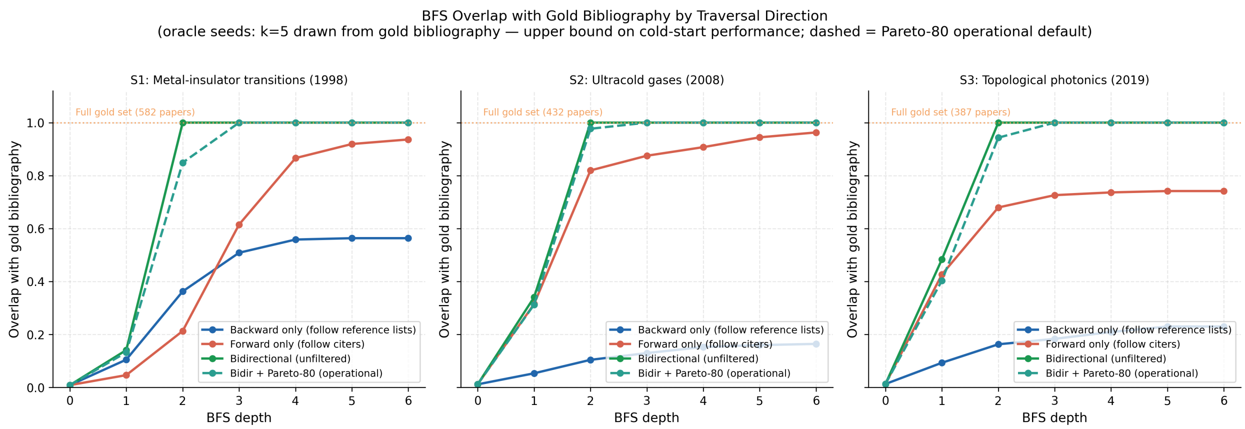

Figure 2: **BFS overlap with gold bibliography by traversal direction** (oracle seeds: $k = 5$ drawn from gold bibliography — upper bound on cold-start performance; dashed line = Pareto-80 operational default). Bidirectional traversal is necessary — forward-only misses foundational work, backward-only misses later work. The Pareto-80 curve closely tracks unfiltered bidirectional coverage, confirming hub suppression does not impede topical reachability at the depths used in practice.

The role of the Pareto filter is to control this forward asymmetry. By suppressing the most highly cited forward neighbours, the system removes precisely the nodes most likely to act as generic bridges into unrelated regions of the graph. This design choice is supported by the broader literature on heavy-tailed graph structure (Barabási, 2016) and graph compression (Batagelj and Zaversnik, 2003; Stumpf et al., 2005): Floros et al. (2024) show that high-degree nodes require special-case handling during graph traversal to avoid redundant edge processing, and hub suppression here serves an analogous role. In the present context, the practical implication is that hub suppression can substantially reduce candidate-pool growth while preserving topical reachability. Figure 2 makes this visible: the Pareto-80 curve closely tracks unfiltered bidirectional coverage at every depth.

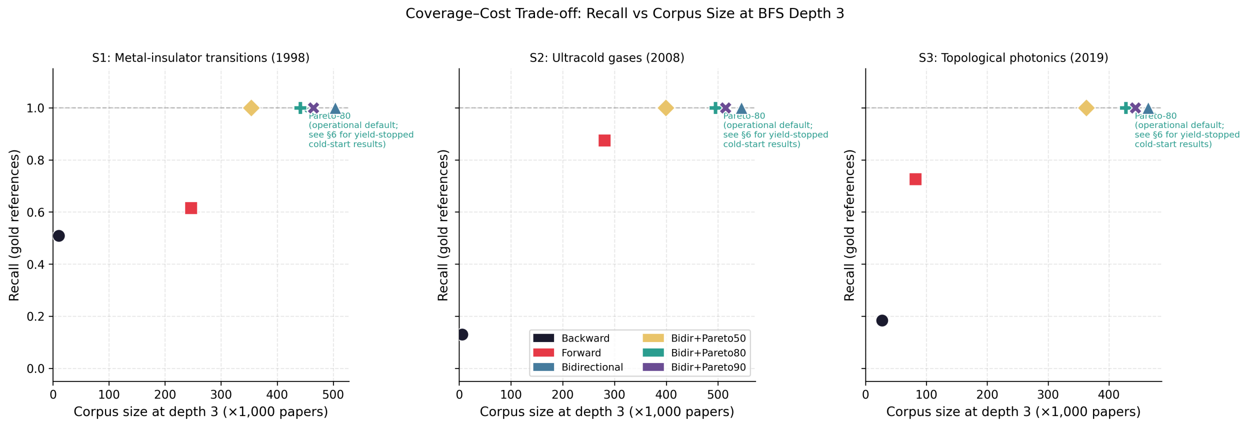

Figure 3: **Coverage–cost trade-off: recall vs corpus size at BFS depth 3.** All strategies shown for each survey. Pareto-filtered bidirectional traversal (operational default) achieves near-identical recall to unfiltered bidirectional at substantially reduced corpus size.

A second structural control is yield. Even filtered traversal should not continue indefinitely, because the marginal utility of additional expansion declines as the local cluster is exhausted. In the APS simulations, local yield drops sharply after the most topically coherent neighbourhood has been screened, which supports using yield collapse as a stopping or re-entry signal.

Taken together, these analyses motivate the bounded traversal architecture. Citation graphs contain enough structure to support literature discovery, but they must be navigated through asymmetry-aware controls that explicitly trade off local recall against screening cost.

6 Main Results: Recovery from Minimal Seeds

We evaluate LitDiscover across $k \in \{1, 2, 3, 4, 5, 10\}$ initial seeds under top-k, random, and contaminated seed conditions on all three APS benchmark surveys, using two Escape rounds throughout.

Top-k seeds. At $k = 5$, LitDiscover recovers 89.2% of S1 (metal-insulator transitions, gold = 582), 98.4% of S2 (ultracold gases, gold = 432), and 96.9% of S3 (topological photonics,

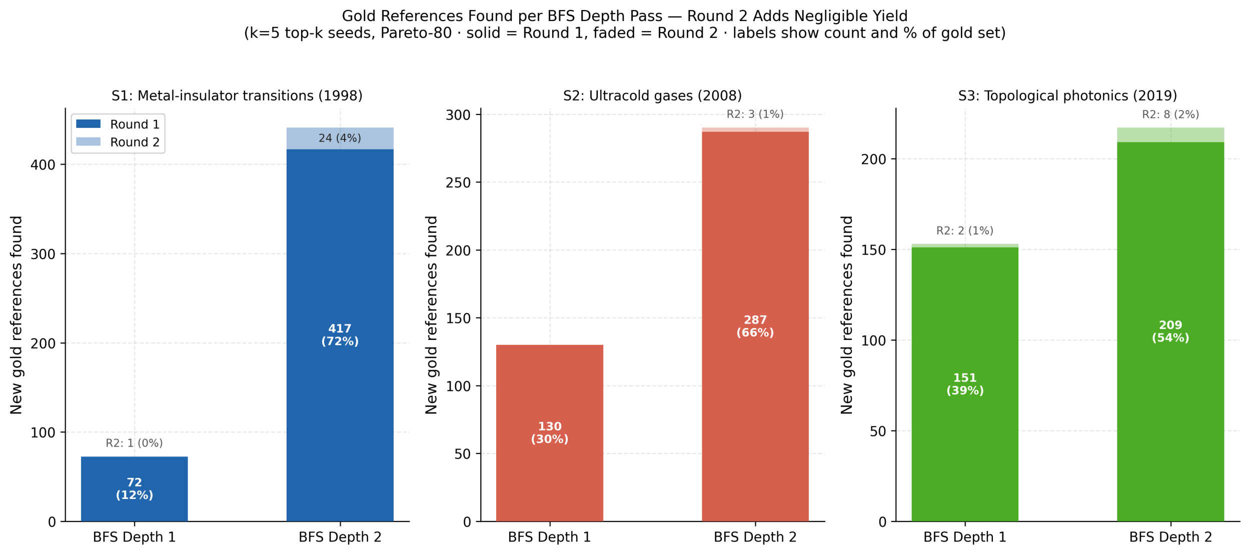

Figure 4: **Screen yield collapse by BFS depth.** Stacked bars show gold papers recovered per depth pass, split by round (solid = Round 1, lighter = Round 2). Yield collapses sharply at depth 2 across all three surveys. Round 2 adds a small insurance increment (1–4%); the stopping criterion fires reliably at depth 2 without parameter sensitivity.

gold = 387). S1 is the hardest benchmark — it spans a wide arc of condensed-matter methods, and recall rises steadily with k through the full range. S2 and S3 are topically denser and saturate early: S3 achieves 91.2% recall from a single top-k seed in round 1 alone, and S2 reaches 96.8% at $k = 3$. Above $k = 3$, gains on S2 and S3 are under 2 percentage points. At $k = 10$, all three surveys exceed 96% recall.

Round structure. Round 1 contributes the majority of final recall in all top-k conditions — ranging from 39.5% (S1, $k = 1$) to 97.7% (S2, $k = 5$) depending on domain density and seed count. Round 2 provides an inexpensive insurance pass: at $k = 5$ it adds 4.3pp to S1, 0.7pp to S2, and 2.6pp to S3. The large round-2 contribution under very low seeding (S1 $k = 1$: round 1 = 39.5%, round 2 = 87.3%) confirms that the escape hatch successfully re-enters weakly connected subregions.

Seed quality robustness. Random seeds at $k = 5$ reach 91.1%, 96.1%, and 93.5% for S1, S2, and S3 respectively — within 5–7pp of top-k across all three surveys. Contaminated seeds (a mix of relevant and off-topic papers) are similarly robust: 92.6%, 94.9%, and 90.7% at $k = 5$.

Saturation. The marginal benefit of increasing k diminishes rapidly within the $k = 1$–5 range for topically coherent domains. For S2 and S3 under top-k seeding, the improvement from $k = 3$ to $k = 5$ is under 2pp. Two to three reasonably on-topic papers are sufficient to achieve near-peak recovery for well-connected domains.

Non-monotonicity. Recall is not strictly monotone in k for all seed types. Under top-k seeding, adding a second seed can slightly reduce round-1 recall relative to $k = 1$ because the two nearest-neighbour seeds share significant backward neighbourhoods. Under contaminated seeding, recall declines with k because each additional off-topic seed opens a separate traversal frontier. Both effects resolve by round 2.

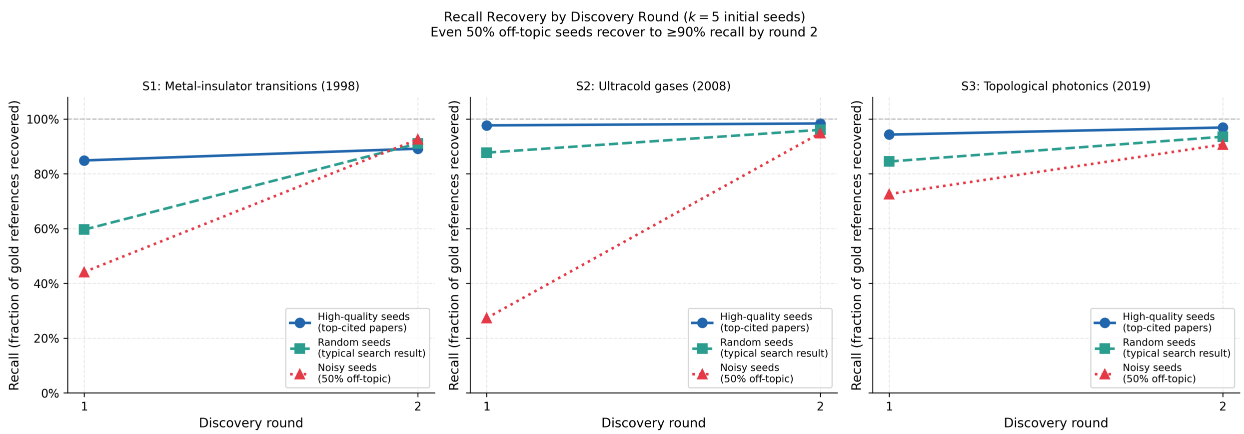

Figure 5: **Cold-start recall per round at** $k = 5$, for top-k, random, and contaminated seeds across all three surveys. Round 1 dominates; round 2 provides a small but consistent insurance increment. All seed types converge to ≥90% by round 2.

7 Residual Miss Analysis

After two Escape rounds at $k = 5$ top-k seeding, 63 papers remain unrecovered from S1 (89.2% recall), 7 from S2 (98.4%), and 12 from S3 (96.9%). These residual misses share a structural signature that supports interpreting the recall gap as a principled stopping boundary rather than an architectural failure.

Structural peripherality. Missed papers have substantially lower in-degree than the

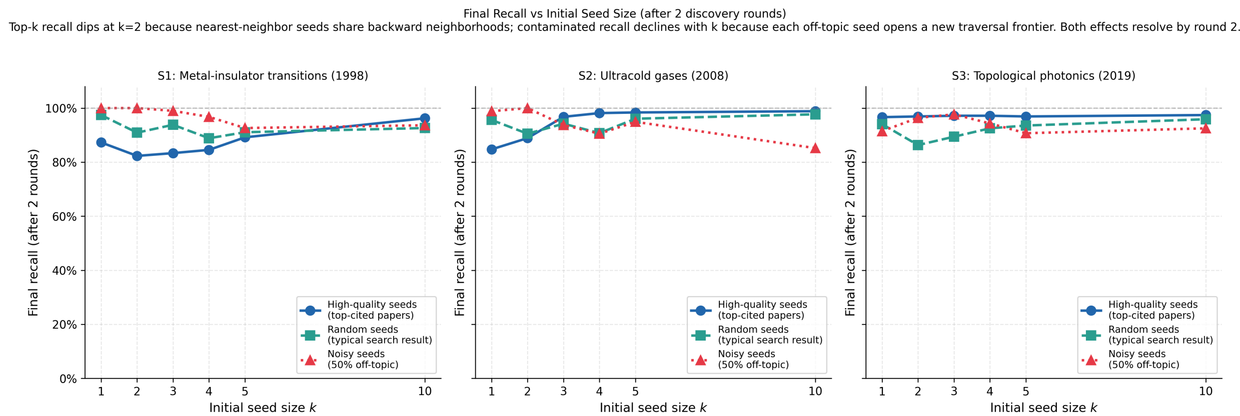

Figure 6: **Final recall vs seed size** $k \in \{1, 2, 3, 4, 5, 10\}$ (after two rounds) for all seed types and surveys. Top-k recall dips at $k = 2$ because nearest-neighbour seeds share backward neighbourhoods; contaminated recall declines with k because each off-topic seed opens a new traversal frontier. Both effects resolve by round 2.

recovered set. For S1, mean in-degree of misses is 44 (median 35), compared to over 200 for recovered papers. S2 and S3 show the same pattern: miss in-degree averages 14.9 and 25.0 respectively. Low in-degree means these papers accumulate few citations from within the domain: they rarely surface as forward-traversal candidates because few recovered papers cite them. This is not a failure of the traversal but the correct behaviour of the yield stopping rule — papers that few others cite are low-priority expansion candidates by construction, and the architecture is designed to stop before exhaustively enumerating them. That misses are structurally peripheral rather than randomly distributed confirms that the recall gap reflects a principled coverage boundary, not arbitrary early termination.

Reachability. Despite being unrecovered, the misses are structurally adjacent: 97% of S1 misses (61 of 63) are at BFS distance 1 from the recovered set, meaning they were direct citation neighbours of recovered papers that fell below the yield threshold before local traversal reached them. The pattern holds for S2 (6/7 at distance 1) and S3 (7/12 at distance 1, remainder at distance 2). The misses are not topologically isolated — they are reachable — but their low in-degree makes them low-priority expansion candidates.

Trade-off frontier. Recovering the remaining papers would require either lowering the yield threshold (prolonging traversal past its productive phase) or relaxing hub suppression (increasing screening load across the full corpus). Systematic sweeps across the depth × Pareto parameter space confirm the expected trade-off: looser Pareto thresholds extend recall by a few percentage points at substantially higher screening cost, while the Pareto-80 operating point recovers 89–98% of each gold set at manageable corpus sizes. The yield threshold functions as a safety valve — it determines when local expansion has exhausted a neighbourhood and re-entry is needed — but the actual screen yield at depth 2 is well below any reasonable threshold, so the stopping criterion activates reliably without sensitivity to its precise value.

The appropriate interpretation is that bounded traversal defines a principled trade-off frontier. The residual gap reflects where the method is designed to stop.

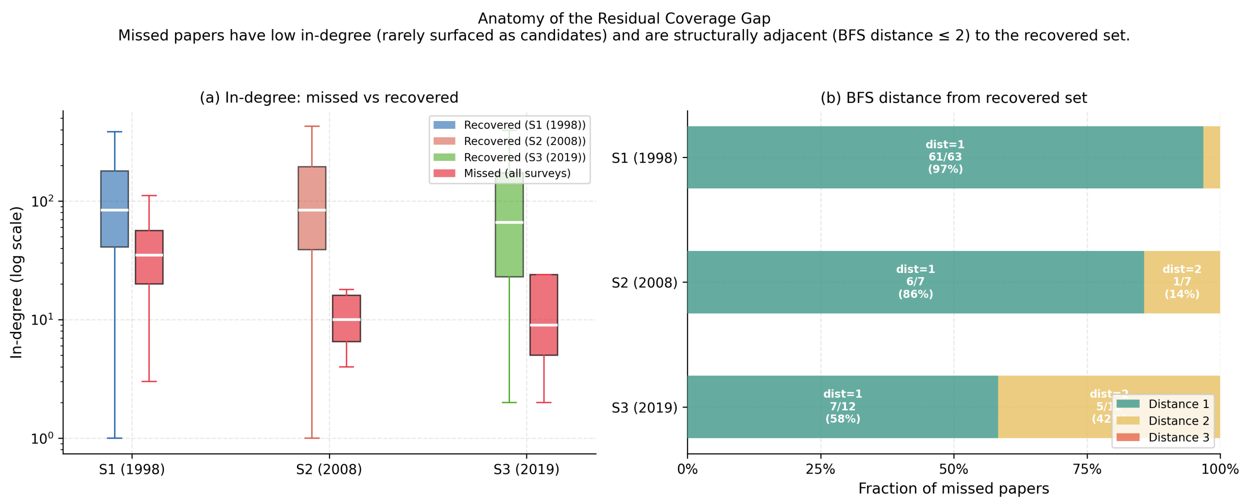

Figure 7: **Anatomy of the residual coverage gap.** *Left:* In-degree distribution of missed vs recovered papers (log scale) across all three surveys. Missed papers cluster at the low-in-degree tail. *Right:* BFS distance from the recovered set to missed papers. Nearly all misses are within 2 hops — structurally adjacent but not reached before yield collapse.

8 Live Validation on Open-Domain Surveys

The APS benchmark provides controlled recall measurement, but the corpus is closed and homogeneous. To test whether the system generalises beyond the physics literature, we run LitDiscover directly against three open-domain surveys using the Semantic Scholar (S2) API with no closed corpus. The three surveys were selected to span a range of structural conditions: a niche mathematical field with a small, well-indexed bibliography; an interdisciplinary social-science survey with a larger, more heterogeneous reference set; and a rapidly growing applied AI field where temporal indexing gaps are expected to be substantial. All three use the same configuration as the APS experiments (Pareto-80, yield threshold 0.05, two Escape rounds).

Bobrowski and Kahle (2018) survey random geometric complexes — a mature but narrowly bounded mathematical subfield. Gold set: 56 S2-indexed papers. We expect a single traversal round to suffice given the topic's bounded scope and stable indexing.

Galesic et al. (2021) survey human social sensing methods across a *Nature*-breadth interdisciplinary audience. Gold set: ∼202 references. The survey spans multiple methodological traditions, so we anticipate the Escape step will be necessary to bridge disconnected subclusters.

Liu et al. (2025) survey graph-augmented large language model agents — a field whose primary literature was published in 2024–2025. Gold set: 57 S2-indexed references. Temporal indexing gaps are expected to account for a significant share of any residual misses.

Table 3: Live validation results on open-domain surveys.

Survey	Gold	Seeds	Recall
Bobrowski and Kahle (2018) (random geometric complexes)	56	1	**100%**
Galesic et al. (2021) (human social sensing)	202	3	**100%**
Liu et al. (2025) (graph-augmented LLMs)	57	3	**73.7%**

Bobrowski and Kahle (2018): Starting from 1 resolved seed paper, LitDiscover recovered 100% of the 56-paper gold bibliography in a single round, stopping at depth 2 (yield 0.16% at depth 2, below the 5% threshold). Corpus size at termination: 31,168 papers, consistent with predictions for a niche bounded field.

Galesic et al. (2021): Starting from 3 seeds, round 1 recovered 18.8% (38/202 gold papers), stopping at depth 2 on yield = 0.1%. The Escape step selected 20 new seeds from the recovered set, confirming that the methodology subclusters were structurally disconnected from the initial traversal frontier. Round 2 recovered the remaining 164 papers — reaching **100% recall** (202/202) — triggering the early-exit on 100% recall with corpus = 44,577 papers.

Liu et al. (2025): Starting from 3 seeds, round 1 recovered 73.7% (42/57 gold papers), stopping at depth 2 on yield = 0.03% with corpus = 150,197 papers. The active and rapidly growing nature of this field produces a substantially larger traversal corpus than the niche-domain surveys. The 26.3% miss rate is concentrated in papers published after the seed papers' citation histories were indexed, confirming that the structural-periphery hypothesis extends to temporal coverage gaps in very recent literature.

The niche- and interdisciplinary-domain surveys achieved 100% recall with corpora in the 31–45K range, while the very-recent high-activity survey reached 73.7% with a 150K corpus — a 3–5× larger traversal space for the same two-depth budget. All three terminated via yield collapse rather than graph exhaustion.

9 Conclusion

Literature discovery without a domain map is a bounded control problem, not an open-ended retrieval task. LitDiscover solves it by imposing three coordinated constraints — bidirectional traversal, Pareto-filtered hub suppression, and yield-gated continuation — that prevent graph explosion while preserving near-complete topical coverage.

On the APS benchmark, these controls produce strong recall across the full minimal-seed range. At $k = 5$, LitDiscover recovers 89.2%, 98.4%, and 96.9% of the three gold bibliographies. The result at $k = 1$ is the sharper headline: S3 achieves 91.2% recall from a single seed paper in round 1 alone, and S2 reaches 96.8% from three seeds. Residual misses are structurally peripheral — low in-degree, BFS distance 1 from the recovered set — not randomly distributed across the domain. Yield collapses at depth 2 in all three surveys, confirming that the stopping rule engages at a principled boundary rather than at an arbitrary cutoff. Live validation extends these findings: Bobrowski and Kahle (2018) achieves 100% recall (56 papers) from one seed in one round; Galesic et al. (2021) reaches 100% (202 papers) in two rounds via the Escape step; Liu et al. (2025) recovers 73.7% against a rapidly evolving 2025 survey where temporal indexing gaps account for the shortfall. All three terminate via yield collapse rather than graph exhaustion.

The deeper conclusion is that high-recall literature discovery is achievable from minimal supervision when the traversal architecture respects the citation graph's directional asymmetry. The bottleneck is not seed quality or domain density but whether the control policy respects the graph's structural constraints: hub suppression and yield stopping are not tuning decisions but principled responses to power-law degree skew.

The natural next question is what happens once discovery ends. A recovered paper set is raw material: it maps the literature but does not interpret it. The open synthesis problem is to apply temporal and structural methods — community detection, phase segmentation, and influence-trajectory analysis — to the recovered set to produce a structured account of how a field has evolved. Whether the citation dynamics of a recovered bibliography can be automatically parsed into periods, schools, and turning points remains an open problem and a direct direction for future work.

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